

Solving Quadratics — Kenny's Method (Completing the Square Proof + Method Selection)

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Standard: CCSS.MATH.CONTENT.HSA-REI.B.4 — Solve quadratic equations in one variable. Derive the quadratic formula by completing the square.

§ 1. Why Kenny's Way? "Add & Subtract Inside" vs "Add to Both Sides"

When you complete the square on $ax^2 + bx + c = 0$ where $a \neq 1$, two methods exist. They both work mathematically, but one is **much safer** for students.

⚠ The Common Mistake (textbook way): Students factor out a , then "add the magic number to both sides" — but they **forget that a is still multiplying the bracket**. So they add $\frac{b^2}{4a^2}$ to the right side instead of $a \cdot \frac{b^2}{4a^2} = \frac{b^2}{4a}$. The equation is now broken, and Evelyn won't know why her answer is wrong.

Example of the danger:

$3(x^2 + 2x) = 5$. Inside, the magic number is 1.

✗ WRONG (both-sides method): $3(x^2 + 2x + 1) = 5 + 1$. (*Forgot to multiply 1 by 3. Right side should be $5 + 3$.*)

✓ RIGHT: $3(x^2 + 2x + 1) = 5 + 3$.

✓ Kenny's Way (Add & Subtract Inside): Don't touch the right side. Inside the bracket, add the magic number AND subtract it — the bracket is mathematically unchanged (you added zero). Then peel out the $-$ piece, multiplying by a as you take it out.

Why this is safer: The leading coefficient a stays attached to whatever leaves the bracket, so you *can't forget to multiply it*.

§ 2. Completing the Square — Full Proof of the Quadratic Formula

Start with the general form. Use Kenny's "add & subtract inside" technique step by step. By the end you will have *derived* the quadratic formula yourself.

Start $ax^2 + bx + c = 0, \quad a \neq 0$

Step 1 Factor a from the first two terms only:

$$a\left(x^2 + \frac{b}{a}x\right) + c = 0$$

Step 2 Magic number = $\left(\frac{1}{2} \cdot \frac{b}{a}\right)^2 = \frac{b^2}{4a^2}$. **Add and subtract it INSIDE** the bracket (net change = 0):

$$a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2}\right) + c = 0$$

Step 3 Peel the $-$ piece OUT of the bracket. Multiply it by a as it leaves:

$$a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}\right) - a \cdot \frac{b^2}{4a^2} + c = 0$$

$$a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}\right) - \frac{b^2}{4a} + c = 0$$

Step 4 The bracket is now a perfect square: $\left(x + \frac{b}{2a}\right)^2$.

$$a\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a} - c$$

Step 5 Common denominator on the right: $\frac{b^2}{4a} - c = \frac{b^2 - 4ac}{4a}$. Then divide both sides by a :

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Step 6 Square root both sides (don't forget $\pm$):

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

Step 7 Subtract $\frac{b}{2a}$ from both sides:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\textbf{Quadratic Formula: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Big idea: The quadratic formula is *nothing but* "completing the square done once for the general case." If you can do completing the square, you already know where the formula came from.

§ 3. Three Methods — Which One When?

Always start by computing the **discriminant** $D = b^2 - 4ac$. The shape of D (and the parity of b) tells you which method is cleanest.

Decision Tree:

1. Compute $D = b^2 - 4ac$.
2. If D is a **perfect square** (0, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, ...) → **FACTOR**. (*Easiest, no radicals.*)
3. Else if b is **even** → **COMPLETE THE SQUARE**. (*Half of b is a clean integer — no fractions.*)
4. Else → **QUADRATIC FORMULA**. (*Always works.*)

⚠ Note: If b is even, the quadratic formula will reduce at the end (the 2's cancel). So completing the square gets you there with cleaner arithmetic. Save the formula for when b is odd or $a \neq 1$ in a messy way.

Method 1 — Factor

Use when $D = b^2 - 4ac$ is a perfect square. Roots are rational.

Find two numbers whose product is ac and whose sum is b . Split the middle term, factor by grouping (or use the standard "two binomials" form when $a = 1$).

Method 2 — Complete the Square (Kenny's "Inside" Style)

Use when D is NOT a perfect square AND b is even.

If $a = 1$: just take half of b , square it, add & subtract inside.

If $a \neq 1$: factor a first, then add & subtract inside the bracket as in § 2.

Method 3 — Quadratic Formula

Use when b is odd, or when $a \neq 1$ and the bracket gets messy.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \text{ Always works, but most arithmetic. Simplify the radical at the end.}$$

§ 4. Worked Examples — One Per Method

Example 1 — FACTOR (D is a perfect square)

Solve: $x^2 - 2x - 15 = 0$

Step 1 (decision): $D = (-2)^2 - 4(1)(-15) = 4 + 60 = 64 = 8^2$. $\Rightarrow$ **Perfect square** $\rightarrow$ **FACTOR**.

Step 2: Two numbers with product -15 and sum -2 : -5 and $+3$.

Step 3: $(x - 5)(x + 3) = 0$.

Answer: $x = 5$ or $x = -3$.

Example 2 — COMPLETE THE SQUARE (b even, D not perfect square)

Solve: $x^2 + 6x + 4 = 0$

Step 1 (decision): $D = 36 - 16 = 20$ (not a perfect square). $b = 6$ is even. $\Rightarrow$ **COMPLETE THE SQUARE**.

Step 2 (Kenny's inside method): Magic number $= \left(\frac{6}{2}\right)^2 = 9$. Add & subtract inside:

$$x^2 + 6x + 9 - 9 + 4 = 0.$$

Step 3: Group: $(x + 3)^2 - 5 = 0 \Rightarrow (x + 3)^2 = 5$.

Step 4: $x + 3 = \pm\sqrt{5} \Rightarrow x = -3 \pm \sqrt{5}$.

Cross-check with formula: $x = \frac{-6 \pm \sqrt{20}}{2} = \frac{-6 \pm 2\sqrt{5}}{2} = -3 \pm \sqrt{5}$. ✓ Same answer, but completing the square skipped the reducing step.

Example 3 — COMPLETE THE SQUARE with $a \neq 1$ (Kenny's "Inside" Style)

Solve: $3x^2 - 2x - 3 = 0$

Step 1 (decision): $D = 4 + 36 = 40$ (not perfect square). $b = -2$ is even but $a = 3 \neq 1$. Try Kenny's inside method (or quadratic formula).

Step 2: Factor out 3 from the first two terms: $3\left(x^2 - \frac{2}{3}x\right) - 3 = 0$.

Step 3: Magic number $= \left(\frac{1}{2} \cdot \frac{2}{3}\right)^2 = \frac{1}{9}$. Add & subtract inside: $3\left(x^2 - \frac{2}{3}x + \frac{1}{9} - \frac{1}{9}\right) - 3 = 0$.

Step 4: Peel the $-\frac{1}{9}$ out (multiply by 3): $3\left(x - \frac{1}{3}\right)^2 - \frac{1}{3} - 3 = 0$.

Step 5: $3\left(x - \frac{1}{3}\right)^2 = \frac{10}{3} \Rightarrow \left(x - \frac{1}{3}\right)^2 = \frac{10}{9}$.

Step 6: $x - \frac{1}{3} = \pm \frac{\sqrt{10}}{3} \Rightarrow \mathbf{x} = \frac{1 \pm \sqrt{10}}{3}$.

Example 4 — QUADRATIC FORMULA (b odd)

Solve: $2x^2 + 3x - 1 = 0$

Step 1 (decision): $D = 9 + 8 = 17$ (not perfect square). $b = 3$ is odd. $\Rightarrow$ **QUADRATIC FORMULA.**

Step 2: $x = \frac{-3 \pm \sqrt{17}}{2(2)} = \frac{-3 \pm \sqrt{17}}{4}$.

Answer: $\mathbf{x} = \frac{-3 \pm \sqrt{17}}{4}$.

Example 5 — The "Reduce at the End" Trap (when b is even but you used the formula)

Solve: $x^2 + 4x - 1 = 0$ — using the formula, then verify by completing the square.

Formula: $x = \frac{-4 \pm \sqrt{16 + 4}}{2} = \frac{-4 \pm \sqrt{20}}{2} = \frac{-4 \pm 2\sqrt{5}}{2} = -2 \pm \sqrt{5}$. (Two reducing steps.)

Completing the square (cleaner): Magic = 4. $x^2 + 4x + 4 - 4 - 1 = 0 \Rightarrow (x + 2)^2 = 5 \Rightarrow x = -2 \pm \sqrt{5}$. ✓

Lesson: When b is even, completing the square gives you the simplified answer in one shot. The formula gives the same answer but you must reduce.

§ 5. Quick Reference Card

Step	What to Check	Method
1	Is the equation set $= 0$?	If not, move everything to one side first.
2	Compute $D = b^2 - 4ac$.	—
3	Is D a perfect square?	YES $\rightarrow$ Factor (or split-the-middle).
4	If no, is b even?	YES $\rightarrow$ Complete the square (Kenny's inside method).
5	Otherwise.	Quadratic formula.

✓ **Habit to build:** Before you start solving, **circle the discriminant** on your paper and write the chosen method next to it. Two-second decision = no wasted minute on a method that will reduce at the end anyway.